Fiscal Year:	FY 2020	**Task Last Updated:**	FY 06/30/2020
PI Name:	Marquez, Jessica J. Ph.D.		
Project Title:	Design of “Autonomous Medical Response Assistant” Interface Prototype for Long Duration Spaceflight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94035	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	TRISH--Synergy
Start Date:	08/17/2019	**End Date:**	02/29/2020
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Menon, Jayant M.D. (Nahlia Inc.) Yashar, Melodie M.A. (San Jose State University Research Foundation)		
Grant/Contract No.:	NNX16AO69A-SYN0008		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Synergy Project NOTE: Continues "Design of 'Autonomous Medical Response Assistant' Interface Prototype for Long Duration Spaceflight--PI Sharpe" as Principal Investigator changed to Jessica Marquez, Ph.D., from previous PI Matthew Sharpe, Ph.D. This project aims to develop a user-centered research and design process to define the needs of crewmembers interacting with an Autonomous Medical Response Assistant (Dr. Menon's project) in long duration spaceflight. The human-centered design augmentation research will entail: domain-specific and analogous research in medical care, interaction design of a screen interface, prototype usability testing and evaluation.		

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Research Impact
Future crewmembers may not have received training in certain medical protocols or it may be years before actually needing to perform a medical task or procedure. The integration of autonomous medical support with vehicle/operations systems promotes confidence between ground support and crew to perform these functions autonomously and without real-time communications. The following represent the core research findings and takeaways relevant to our project, "Design of 'Autonomous Medical Response Assistant' Interface Prototype for Long Duration Spaceflight." Numerous possibilities for advancement of present research are likewise introduced:

Integration Requirement: Our research validated that integration of mission-critical information and vehicle data with autonomous medical decision support technologies is highly critical from a human systems integration perspective. The in-flight medical system should provide comprehensive health management relevant to prevention, diagnosis, treatment, and long-term management of crew health. The on-board medical system should also: minimize crew burden while using the system, share a common user interface with the overall vehicle system to minimize crew training and cognitive burdens, reduce operations complexity, and lower mission medical risk (Mindock et al. 2017).

Crew Autonomy Technologies for medical decision support are critical to augment and enable crew autonomy, but the degree of autonomy will depend on the communication time delay and the training of the crew. The assumed or implied agency of an autonomous medical system such as Autonomous Medical Response Assistant (AMRA) within an emergency scenario where a physician-crewmember cares for a patient-crewmember is particularly under-examined. Exploration Medical Capabilities (ExMC) element rightly specifies the caretaker as a physician, and not just a CMO (Chief Medical Officer). The mission physician-astronaut serves as director of care during real-time medical events and thus acts as the primary source for in-mission medical decision making.

Crew Confidentiality Designing for or assuming crew confidentiality (such as in protecting the privacy of personal health information) in future long duration exposure missions (LDEMs) requires further research and investigation. We found there is insufficient data indicating how crew confidentiality would differ substantially from International Space Station (ISS) medical operations, and additional research is necessary to identify how the role of the mission physician-astronaut will replace or impact the role of the flight surgeon in mission control today.

Communications Protocols Emergency (pre-approved) protocols (such as in medical manuals) will still be extremely critical when time is of the essence. Today aboard ISS, the crew is instructed to go through the Medical Manual checklist and then call a flight surgeon for next steps. The establishment of communications protocols for self-care of an unplanned incident between the patient, on-board medical system, and ground versus directed care through both planned and unplanned incidents with a patient, caretaker, on-board medical system ground is a significant area of research with unique requirements still demanding consideration.

Ground Awareness & Trust: Crew to ground synchronization of medical records (such as the EMR--electronic medical record) and documentation of unplanned incidents will prove challenging and represent an unprecedented paradigm from ISS. Regardless of whether an acute or chronic condition is being diagnosed or treated, information must be sent to flight surgeons on the ground and will remain 'in-the-loop' to adequately and collaboratively manage their health and promote trust between ground and crew. Even though the synchronization of in-mission and ground medical data systems depends on a variety of factors such as telemetry bandwidth, distance from Earth, priority of data, and more, ground support must continue to be informed on the state of the crew to assess impacts to the mission and provide support as needed.

Reference

Mindock J, et al. (2017) Integrating spaceflight human system risk research. Acta Astronautica 139:306-312.

Introduction
An autonomous medical response assistant (AMRA) is envisioned to help astronauts address medical complaints, develop a differential diagnosis, and guide self-treatment until a healthy state is restored. Major challenges for astronauts in future long-duration exploration missions (LDEMs) will be that crewmembers are not expected to be medical professionals, may be under high workload and high stress, are facing physiological challenges caused by spaceflight, and will have limited voice communications with medical support from Earth. AMRA develops a process of personalized diagnosis and treatment through a Bayesian predictive control system that recommends therapeutic control actions including diagnostic tests and treatments to crewmembers (see Menon AMRA record-- https://). In our project, "Design of 'Autonomous Medical Response Assistant' Interface Prototype for Long Duration Spaceflight," a user interface that guides astronauts through self-diagnosis, treatment, and rehabilitation while communicating with remote specialists in ground support (most notably a patient's flight surgeon) is developed and tested in an interactive user-interface prototype. Our project develops the interaction design for the crewmember using AMRA through user research, iterative design, and usability testing to evaluate the user interface and workflow designed.

Original Project Aims / Objectives: An Autonomous Medical Response Agent for future long-duration exploration missions must consider not only decision-making support systems for self-diagnosis of medical complaints from the crew in times when feedback or guidance from ground support is not immediately available, but must also consider recommendations of appropriate treatment protocols which integrate with pre-existing databases of medical manuals for spaceflight and otherwise, most of which have yet to be synthesized and integrated within a proof-of-concept technology interface. The goals of the project were to create a prototype that: 1) provides an optimal work flow of actionable recommendations for self-care of the crew while 2) building trust between Mission Control Center (MCC) flight surgeons and astronauts on long-duration exploration missions (LDEM) with limited bandwidth for communications. The scope of our project focused on the workflow of an astronaut and a caretaker working with AMRA in reviewing, diagnosing, and treating a chief medical complaint. AMRA's algorithms function to monitor the crew's normative or healthy state, provide a means for developing a differential diagnosis, and identify a protocol for diagnostic and treatment response to a medical incident. Envisioned as a component of the on-board medical system in future LDEMs, this same technology interface must communicate and synchronize with human health monitoring from ground support—so that feedback, interventions, and treatments may be communicated appropriately. Despite the fact that communication between the crew and ground will be challenging due to time delays and bandwidth limitations, flight surgeons on the ground will still

Task Progress:	need an effective means to monitor astronaut health for LDEMs (Love & Reagan, 2013). Key Findings The prototype deliverable for this project, titled AMRA Aggregate Information Display (AMRA AID), is an integrated information display system for comprehensive autonomous medical guidance, diagnosis, and treatment of in-flight medical conditions experienced by crewmembers. AMRA AID demonstrates how we might ensure crew autonomy, increase the crew's medical capabilities, and decrease cognitive burden within a front-end user interface. This project arrives at critical findings regarding usability needs, communication requirements, and integrated information requirements for a future technology interface functioning to increase confidence between ground support and LDEM crewmembers. AMRA AID's comprehensive view enables efficient information access for both crew & ground support, reducing cognitive burden in the event of an unplanned or emergency medical incident and enabling informed analytical decisions to be made based on both crew and vehicle health. Human-centered design augmentation advanced within the web-based prototype included: enhanced workflow and treatment guidance for two medical scenarios for a non-specialist user base with various levels of medical training, interaction design which considered speech (conversational user interface) elements and on-screen interactions to be developed in future iterations of the project, communication design and functional requirements relevant to self-care versus caring for another astronaut, as well as user testing of the prototype with an international space medical community. Impact of Key Findings Based on the two elaborated LDEM medical scenarios, the research demonstrated the need for additional research relevant to information integration, confidentially, and communications to enable trust. The primary risk addressed by this project is the Risk of Adverse Health Outcomes & Decrements in Performance due to Inflight Medical Conditions [Human Research Program-HRP Element ExMC], and the secondary is the Risk of Inadequate Human-Computer Interaction (Short Title: HCI) [HRP Element Human Factors & Behavioral Performance-HFBP)]. The two medical scenarios developed in this project—headache and shortness of breath—have been identified as high impact conditions within the Exploration Medical Conditions List (EMCL) and develop subsequent work on the scenarios by ExMC to develop functional requirements for the on-board medical system. Our scope within this project was refined to demonstrate diagnosis, treatment, and management of long-term patient care in addition to demonstrating the requirement (from a human systems integration perspective) that the front-facing user interface for the medical system integrate mission-critical information from flight sub-systems such as vehicle information and crew schedule. Reference Love SG, Reagan ML (2013) Delayed voice communication. Acta Astronautica 91:89-95.
Bibliography Type:	Description: (Last Updated: 05/06/2026)
Abstracts for Journals and Proceedings	Yashar M, Torron I, Marquez JJ, Menon J, Joyce R, Sharpe M. "Design of Autonomous Medical Response Agent (AMRA) Aggregate Information Dashboard (AID)." 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. Abstracts. 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. , Jan-2020